

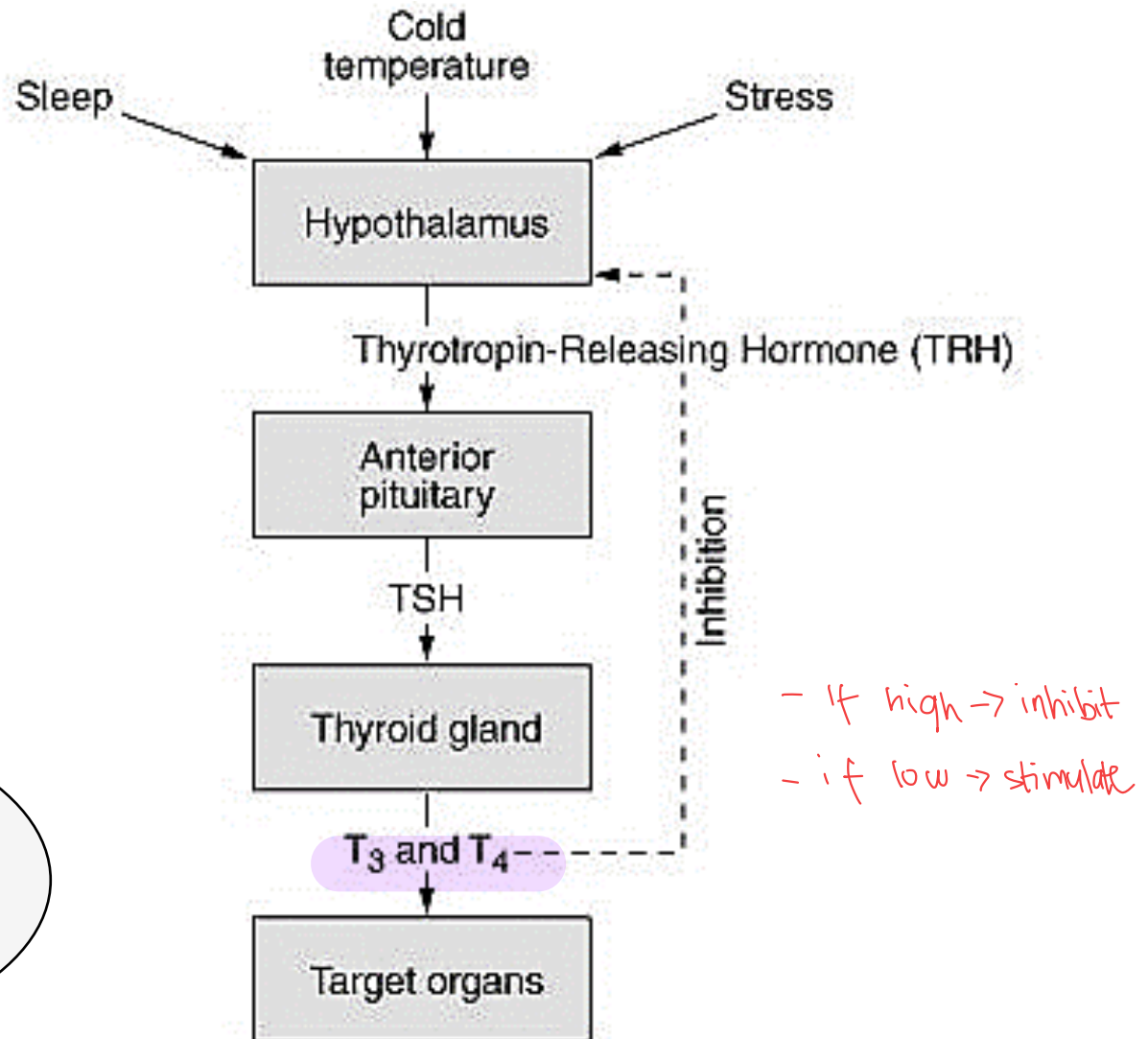
PATH 1017

Thyroid Disorders

Professor Jennifer Lamarre

Thyroid Hormone Regulation Review

- Pituitary-hypothalamus axis and negative feedback mechanism



Do you remember the various thyroid hormones from anatomy class?

Thyroid Diagnostic Tests

- Blood tests:
 - **TSH**
 - **T3 and T4**
 - Serum-free T4
 - T4 resin uptake
 - **Thyroid antibodies**
- Radioactive iodine uptake
- Fine-needle biopsy
- Thyroid scan, radio scan, or scintiscan

Why is TSH tested first?

What is the purpose of the biopsy?

Thyroid Imbalances

Hypothyroidism

- **Congenital** (present at birth)
- **Acquired**
 - Hashimoto thyroiditis
 - Thyroidectomy

Hyperthyroidism

- Graves disease
- Thyroid tumors

Goiter

Associated with:

- 1) Over-secretion of thyroid hormone (hyperthyroidism)
- 2) Iodide deficiency leading to inadequate thyroid hormone levels (hypothyroidism)

- Why do goiters occur?

- Why is it rare to see goiters in North America?

Acquired Hypothyroidism

- Suboptimal thyroid function
- Develops later in life, women 5x>men
- Leads to slowing of metabolism due to decreased thyroid hormone
- Severity varies from mild to life-threatening

Acquired Hypothyroidism

Caused by:

- Disease or destruction of thyroid gland (primary hypothyroidism)
 - **Hashimoto thyroiditis** (most common) –what happens?
- Hypothalamus or pituitary gland disorders (secondary hypothyroidism)
- Other-
 - medications (lithium)
 - ingestion of large amounts of iodide- what happens?
 - hyperthyroidism treatments

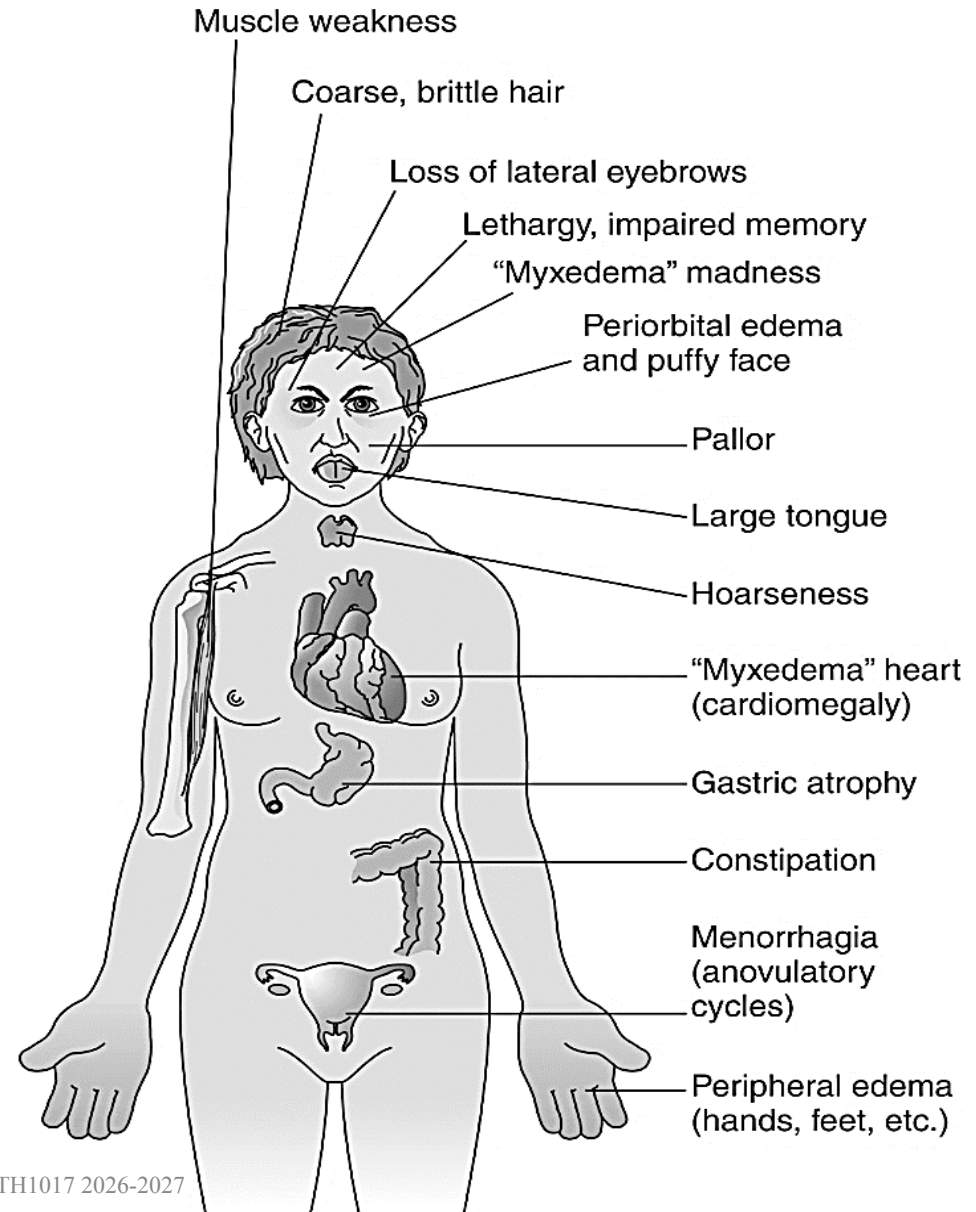
Hypothyroid Manifestations

- Vary from mild to severe
- Caused by:
 - Hypometabolic state due to thyroid hormone deficiency
 - Myxedematous involvement of body tissues
 - Hard non-pitting edema caused by increased quantities of mucins in subcutaneous tissue that trap water in interstitial space
- Can affect organs and leads to severe signs

Hypothyroidism Manifestations

Which manifestations are most common?

Which manifestations occur early vs. later in the illness?



Hypothyroidism - Diagnostics

Lab values:

-
-
-

Medical Management of Hypothyroidism

- Synthetic levothyroxine-replacement therapy
- Be aware of medication interactions
 - Effects of hypnotic and sedative agents: reduce dosage
 - Impact on diabetic patients
- Support of cardiac function and respiratory function
- Prevention of complications (i.e. CAD)

Congenital Hypothyroidism

- Affects 1 in 5000 infants
- Inadequate thyroid hormone secretion during fetal development
- Factors:
 - Lack of thyroid gland
 - Abnormal synthesis of thyroid hormone
 - Too little TSH secretion

The baby often appears normal at birth – why?

What manifestations might the baby have?

Congenital Hypothyroidism

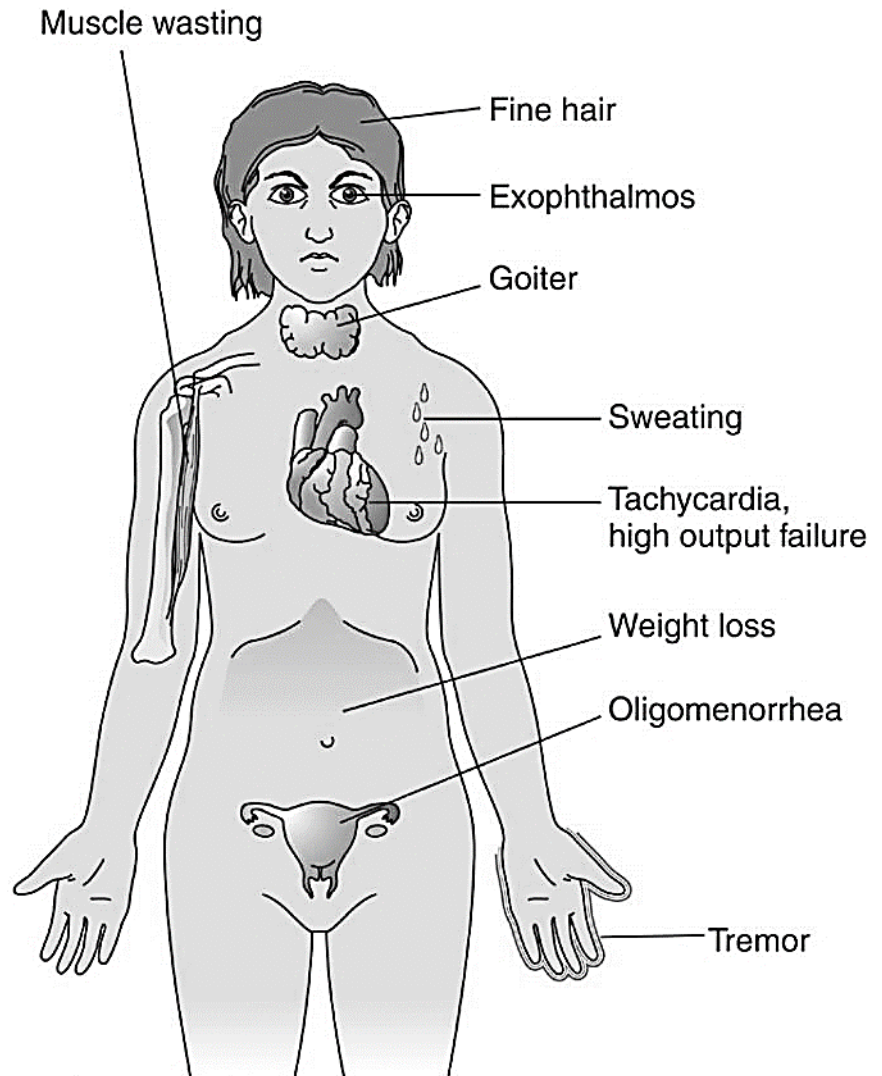
- If untreated- “cretinism” develops –effects?
- Screening Test: heel stick blood test for T4 or TSH
- Treatment- thyroid hormone replacement
- Prognosis- good if begun in first 6 weeks

Hyperthyroidism

- “thyrotoxicosis” occurs due to excessive levels of thyroid hormones
- Caused by:
 - Graves disease (most common)
 - Other-multi-nodule goiter; thyroid adenoma (cancer); excess thyroid hormone ingestion (meds)

Hyperthyroid Manifestations

- Caused by:
 - Increased O₂ consumption & use of metabolic fuels due to hypermetabolism
 - Increased sympathetic nervous system activity
- What are the most common manifestations?



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Graves Disease

- Autoimmune disorder
 - Excessive stimulation of the thyroid gland and thyroid hormone release by antibodies
- Leads to hyperthyroidism accompanied by classic signs:
 - Goiter
 - Exophthalmos
 - Dermopathy (thickening of skin)
- Onset- 20-40 years
- Women 5x > men

Hyperthyroidism - Diagnostics

Lab results:

-
-

Medical Management of Hyperthyroidism

- Radioactive ^{131}I therapy
- Medications:
 - Propylthiouracil and methimazole
 - Sodium and potassium iodine solutions
 - Dexamethasone
 - Beta-blockers
- Surgery and subtotal thyroidectomy

- What is the purpose of radioactive iodine?
- Why beta-blockers?

Thyroid Storm (Crisis)

- **Rare**, extreme, life-threatening form of thyrotoxicosis (++++ T4)
 - Often related to undiagnosed or inadequately treated hyperthyroidism
 - High mortality rate
- Precipitating factors:
 - Manipulation of thyroid gland during surgery (most common)
 - Stress, infection, DKA, trauma, emotional event.
- Manifestations:
 - Very high fever,
 - Extreme CV effects –examples?
 - Severe CNS effects –examples?

Thyroid Storm (Crisis)

- Requires rapid diagnosis and treatment
- Supportive measures:
 - Cooling measures
 - Fluids, glucose and electrolyte replacement
 - Beta-blocker drug (propranolol)
 - Glucocorticosteroids
 - Propylthiouracil
 - Iodide contrast